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April 19, 2026

The Editors
Foundations of Physics

Dear Editors,

I am submitting **"SIC-POVM Existence via Frobenius-Fixed Stark Units"** for consideration in *Foundations of Physics*.

The paper began as an attempt to explain a coincidence. Two dimensions of the SIC problem — $d = 19$ and $d = 48$, separated by 29 with no obvious arithmetic relationship — returned identical coordinates in a structural type-theoretic framework I have been exploring (open source, unlicensed, and publicly available at `https://github.com/umpolungfish/synthomnicon`). Not similar: identical, modulo a single primitive. My immediate response was to assume I had made an error. I checked; I had not. My second response was to assume the framework was doing something degenerate. I checked that too. What I found instead was that the coincidence was pointing at something real about the structure of the problem, and that following it carefully led somewhere I had not expected.

The twenty-five year impasse on SIC-POVMs is not, I now believe, a computational problem. The numerics are extensive and reliable; the dimensions where SIC-POVMs are known number well over a hundred. The difficulty is that the algebraic self-duality which equiangularity requires — a palindromic Z_2 symmetry in the minimal polynomials of the fiducial coordinates — is not the kind of property that can be approximated into or assembled from weaker ingredients. It has to be present, in full, in the object one starts from. Proving that it is present is what the paper reduces to, and the object that already has it intrinsically is the Stark unit in the associated ray class field.

What I think the paper contributes, beyond the conditional proof itself, is a precise account of why the problem has been hard. That account has a specific prediction: two of the four conjectures on which the theorem rests are new to the literature, and the crux of the argument, a conjecture I label H2a about the eigenspace structure of the Dirichlet regulator matrix, is in my view more tractable than the original geometric conjecture in Hilbert space. Whether that assessment is right is for working number theorists to judge. But stating the problem in that form — rather than bundled under a single "Stark hypothesis" as in earlier treatments — seems to me to reorganize what needs to be done in a useful way.

I am submitting to *Foundations of Physics* because the SIC-POVM existence question is at its core a question about the structure of quantum state space, and that question deserves to be in conversation with the readership of this journal. The approach here is arithmetic rather than geometric, and the connection it opens to the Langlands program is not decoration: if the Galois-to-Weyl correspondence in the paper is what I think it is, the quantum phase space group and the arithmetic class group of a real quadratic field are not merely analogous but genuinely dual, in a sense that belongs to the abelian case of the Langlands dictionary. That would make the SIC conjecture a problem in Hilbert's 12th problem as much as in quantum information, and I think *Foundations of Physics* is the right place to put that observation in front of people who have reason to care about both sides of it.

I will add, less modestly: I hope this paper might serve as a small example of the heterodox thinking the field has long, and now desperately, requires. Twenty-five years of numerical evidence for SIC-POVMs is an extraordinary accumulation, and it is reliable. But reliable accumulation is not the same as progress toward a proof, and I think the SIC community has been in a quantitative quagmire long enough that the value of approaching the problem from a direction the computational tradition cannot reach ought to speak for itself. Whether this particular direction is the right one is precisely what peer review is for.

The main theorem is conditional, and I have not softened that fact. Four conjectures remain open. I believe identifying them precisely, separating their logical dependencies, and showing that they are jointly sufficient is a genuine contribution to the problem — but that is, ultimately, for the editors and referees to assess. I appreciate your time, and hope you enjoy the manuscript.

Sincerely,

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